

Market Risk Project Report

ESILV — Market Risk (2025–2026)

Students (TD IF2)

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1 Introduction

This project was carried out in the context of the *Market Risk* course at ESILV (academic year 2025–2026). The point is not only to “compute a VaR”, but to apply the course methods on real data, justify our choices, and then look at the results with a critical eye (including the moments where reality does not behave nicely).

We provide two complementary supports:

- **This report**, which is meant to be self-contained and readable. We focus on the main steps, the key assumptions, and the final outputs. We also include the most relevant code blocks (as requested), but we try to keep them targeted, so the report stays pleasant to read.
- **A Jupyter notebook**, which contains the full pipeline, intermediate checks, and additional plots. If someone wants to reproduce everything line by line (or see where a specific plot comes from), the notebook is the safest place to look.

We will occasionally refer to the notebook when we skip an intermediate step or a technical detail. It is not a way to avoid explanations: it is just our way to keep the report focused and avoid turning it into a long code dump.

The project is structured into five questions (A to E), following the official statement:

- **A:** Non-parametric Value-at-Risk on daily Natixis returns, using a biweight kernel density estimator, and an out-of-sample backtest.
- **B:** Expected Shortfall associated with the VaR from question A, and a practical comparison between both risk measures.
- **C:** Extreme Value Theory: Pickands tail index estimation (both tails), then EVT-based VaR for several confidence levels under an i.i.d. assumption.
- **D:** Calibration of Bouchaud’s price impact model on the TD4 dataset, with a discussion about parameter meaning and model specification.
- **E:** Haar wavelets: multiresolution correlations between FX rates (Epps effect), then Hurst exponent estimation and volatility scaling.

Throughout the report, unless explicitly stated otherwise, we use the standard convention for daily VaR/ES reporting with 95% as the baseline confidence level, and additional levels when it is useful. The exact levels are always easy to change in the code, and we keep the code consistent with the notebook.

2 Question A — Non-parametric Value-at-Risk

2.1 Objective

In this question, we estimate a one-day historical Value-at-Risk (VaR) on Natixis daily returns using a non-parametric approach based on a **biweight kernel**. The probability level is treated as a **parameter** (easy to change). We then test the VaR threshold out-of-sample on the period 2017–2018.

2.2 Data and returns

We start from daily closing prices and compute simple daily returns:

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}}.$$

We work with 2015–2016 for the estimation step and keep 2017–2018 for the out-of-sample validation.

Code (data loading, returns, and split)

```
[language=Python] data = pd.read_csv("data/NatixisStock.csv", sep = ";")
data["Date"] = pd.to_datetime(data["Date"], dayfirst = True) data["Price"] = data["Price"].str.replace(",",".")
data = data.sort_values("Date")
[language=Python] data["Return"] = data["Price"].pct_change() data = data.dropna()
[language=Python] data_2015_2016 = data[(data["Date"] >= "2015-01-01") & (data["Date"] <= "2016-12-31")]
data_2017_2018 = data[(data["Date"] >= "2017-01-01") & (data["Date"] <= "2018-12-31")]
```

2.3 A.1 — Non-parametric VaR estimation (2015–2016)

2.3.1 Kernel choice and KDE

Following the statement, we use the biweight kernel

$$K(u) = \frac{15}{16}(1 - u^2)^2 \mathbf{1}_{|u| \leq 1},$$

and estimate the density of returns with a kernel density estimator (KDE). In practice, the KDE gives us a smooth estimate of the return distribution, which is especially useful when we look at tail quantiles.

Code (biweight kernel, bandwidth, KDE)

```
[language=Python] def biweight_kernel(u): return (15/16) * (1 - u**2)**2 * (abs(u) <= 1)
[language=Python] We focus on the returns observed over the estimation period (2015–2016)
returns = data_2015_2016["Return"].values n = len(returns)
Bandwidth selection using a rule-of-thumb formula h = 1.06 * np.std(returns) * n**(-1/5)
def kde_biweight(x, data, h): u = (x - data) / h return np.sum(biweight_kernel(u)) / (len(data) * h)
```

2.3.2 Visual inspection (quick sanity check)

Before extracting the VaR, we plot the estimated density to check that the shape is reasonable and to get a first visual feel for the left tail.

2.3.3 VaR computation (KDE-based quantile)

The VaR at level α is defined as the α -quantile of the return distribution:

$$\text{VaR}_\alpha = F^{-1}(\alpha).$$

This definition is exactly the one used in the lecture notes.¹

In the notebook, we build an estimated CDF on a grid (numerical integration) and then interpolate the point where the CDF reaches α . This is a simple and robust way to extract the quantile from the non-parametric distribution.

¹Market Risk lecture slides, M. Garcin, p. 68 (as referenced in our notebook).

Code (alpha parameter, KDE-based CDF, VaR)

[language=Python] Risk level (easy to change) $\alpha = 0.05$ 5

We approximate the CDF implied by the KDE by numerical integration on the grid. This is a simple way to extract the alpha-quantile from the non-parametric distribution.

$dx = x_{grid}[1] - x_{grid}[0]$

Normalize the KDE to make sure it integrates to 1 on the grid (numerical stability)

$kde_{area} = np.trapezoid(kde_{vals}, x_{grid})$ $kde_{pdf} = kde_{vals} / kde_{area}$

$kde_{cdf} = np.cumsum(kde_{pdf}) * dx$ $kde_{cdf} = np.clip(kde_{cdf}, 0, 1)$ just to avoid tiny numerical overshoots

VaR threshold = alpha-quantile of the KDE-based distribution $VaR_{nonparametric} = np.interp(\alpha, kde_{cdf}, x_{grid})$

print(f"Non-parametric (KDE-based) VaR threshold : $VaR_{nonparametric} : .6f$ ")

2.3.4 Graphical summary (VaR location)

The following plots make the quantile idea very concrete: we locate the VaR threshold on the KDE density, and we also show the empirical CDF versus the KDE-based CDF.

Code (plots for KDE pdf and CDF comparison)

[language=Python] Before computing the VaR, we plot the estimated density to visually check the shape of the return distribution, especially the left tail. $x_{grid} = np.linspace(returns.min(), returns.max(),$

$kde_{vals} = np.array([kde_{biweight}(x, returns, h) for x in x_{grid}])$

$plt.figure(figsize=(8, 4))$ $plt.plot(x_{grid}, kde_{vals})$ $plt.title("Non-parametric density estimate of returns (2015$

[language=Python] Empirical CDF on the same grid (useful comparison with KDE-based

CDF) $ecdf_{vals} = np.array([np.mean(returns <= x) for x in x_{grid}])$

$plt.figure(figsize=(10, 5))$ $plt.plot(x_{grid}, kde_{pdf}, label = "KDEpdf (Biweight)")$ $plt.axvline(VaR_{nonparametric},$

"--", label = "VaR threshold") $plt.title("Non-parametric KDE and Value-at-Risk (2015~2016)")$ $plt.xlabel("$

$plt.figure(figsize=(10, 5))$ $plt.plot(x_{grid}, ecdf_{vals}, label = "Empirical CDF")$ $plt.plot(x_{grid}, kde_{cdf}, label =$

"KDE-based CDF") $plt.axvline(VaR_{nonparametric}, linestyle = "--", label = "VaR threshold")$ $plt.axhline($

" : ", label = "alpha level") $plt.title("Empirical and KDE-based CDF with VaR location (2015~2016)")$ $plt.xlabel("$

2.4 A.2 — Out-of-sample validation (2017–2018)

We now keep the VaR threshold learned on 2015–2016 and test it on 2017–2018. A violation occurs when the realized return is smaller than the VaR threshold.

In our out-of-sample check, we obtain a violation rate that is clearly below the theoretical level α . In other words, with this split, the VaR looks conservative for the 2017–2018 regime. This does not necessarily mean the method is wrong: it mostly highlights that market conditions can shift from one period to another (volatility and tail behavior are not constant over time).

Code (backtest and violation rate)

[language=Python] Question A-b: out-of-sample check (2017–2018) $returns_{out} = data_{2017:2018}["Return"].values$

$violations = returns_{out} < VaR_{nonparametric}$ $n_{obs} = len(returns_{out})$ $n_{viol} = int(np.sum(violations))$ $viol_{rate} = n_{viol} / n_{obs}$

print("Out-of-sample backtest (2017–2018)") print(f"Observations: n_{obs} ") print(f"Violations (return < VaR) : n_{viol} ") print(f"Observed violation rate : $100 * viol_{rate} : .2f$ ") print(f"Theoretical level (alpha) : $100 * \alpha : .2f$ ")

2.5 Short wrap-up

Question A gives us a clean baseline: a fully non-parametric VaR using a biweight KDE (2015–2016) and a direct out-of-sample validation (2017–2018). We keep the approach simple, transparent, and reproducible, with α as a parameter that can be changed in one line.

[width=0.75]figures/A1 $_k$ *de_density*₂₀₁₅₂₀₁₆.png

Figure 1: Non-parametric density estimate of returns (2015–2016) using a biweight KDE.

[width=0.85]figures/A2 $_k$ *de_var*₂₀₁₅₂₀₁₆.png

Figure 2: KDE density (biweight) with the VaR threshold location (2015–2016).

[width=0.85]figures/A3 $_cdf$ *comparison*₂₀₁₅₂₀₁₆.png

Figure 3: Empirical CDF vs KDE-based CDF, with VaR location and α level (2015–2016).